

Tutorial 1

Matrices

1. *In mathematics you have to take definitions very seriously. They give you **all** the information about a new concept. If you want to use the new concept, the definition is the only information you have (at least until you have proved a few theorems). Learning to work with definitions is a very important mathematical skill. This question is designed to help you develop this skill.*

Read the following definitions carefully (a couple of them are not new):

The *transpose* of an $n \times n$ matrix A is the matrix A^T obtained from A by interchanging the rows and columns of A .

An $n \times n$ matrix $A = (a_{ij})$ is called:

upper triangular if $a_{ij} = 0$ for all $i > j$,

diagonal if $a_{ij} = 0$ for all $i \neq j$,

symmetric if $A^T = A$,

skew-symmetric if $A^T = -A$.

Now do the following questions quickly.

- (a) Give an example of a 3×3
(i) upper triangular matrix (ii) diagonal matrix (iii) skew-symmetric matrix.
- (b) How do you think a **lower triangular matrix** should be defined? Write out your definition carefully, and then give an example of a 3×3 lower triangular matrix.
- (c) Complete: If a square matrix A and its transpose are both upper triangular, then A is a matrix.
- (d) Complete: A symmetric lower triangular matrix is a matrix.
- (e) For each of the following statements, say whether it is true or false. Give a reason for your answer.
- A square matrix cannot be both upper and lower triangular.
 - If A is a square matrix, $A - A^T$ is a skew-symmetric matrix.
 - If A is an upper triangular matrix, then $a_{ij} \neq 0$ for all $i \leq j$.
 - If A is a diagonal matrix, then $a_{ii} \neq 0$ for every i .
 - If A is a diagonal matrix, there is a non-zero vector $\mathbf{x}$ and a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$.
 - If A is an upper triangular matrix, there is a non-zero vector $\mathbf{x}$ and a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$.

2. Decide which of the following matrix products are defined, and where possible find the product.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

3. Find real numbers a, b, c, d, e satisfying the following equations. You should be able to do this just by looking; no calculations required at all.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = a \begin{pmatrix} 1 \\ 4 \end{pmatrix} + b \begin{pmatrix} 2 \\ 5 \end{pmatrix} + c \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

(b) $\begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = d \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + e \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

4. Let $A = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 9 \\ -3 & k \end{pmatrix}$. What value(s) of k , if any, will make $AB = BA$?

5. Let $A = \begin{pmatrix} 3 & -6 \\ -2 & 4 \end{pmatrix}$. Construct a 2×2 non-zero matrix B such that AB is the zero matrix.

6. Prove that for any $m \times n$ matrix A and any $n \times r$ matrix B ,

$$(AB)^T = B^T A^T,$$

where A^T stands for the transpose of A .

[Hint: Prove that the entry in row i , column j of $(AB)^T$ is equal to the element in row i column j of $B^T A^T$.]

7. A square matrix P is called a *stochastic*, or *Markov matrix* if all its entries are non-negative and the entries in each column add up to 1. Prove that the product of two stochastic matrices is again a stochastic matrix.